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Fluidized Bed Boilers

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ABSTRACT

Fluidized bed combustion has the capability of burning low-grade fuels effectively. These fuels include wood, wood wastes and agrobiofuels, as well as demolition wood and recycled fuels. However, biofuel combustion has occasionally suffered e.g. from bed material sintering and fouling of superheaters and other cooling surfaces. These symptoms have restricted the wide utilization of new biofuels and energy fractions of material recycling.

The fouling tendency of fuels is not dependent on their ash content only. More important factors are the composition of the ash formed in the combustion and the ash mineral reactions. The mechanisms of the deposit formation, origin of ash compounds and their vaporization were studied in cooperation with the Technical Research Centre of Finland (VTT) and Åbo Akademi University.

Deposits on tube heat exchangers were collected by using a temperature controlled deposit probe. Short probe tests were used for studying the deposit at the initial state of formation. The formation mechanisms of ash particles and deposits were studied with measurements of fly ash particle size distribution by VTT. The impaction of partly molten particles is anticipated to cause drastic fouling. Åbo Akademi University studied the ash reactions with their novel ash predictor, where the elementary composition of the fuel ash was used for equilibrium calculations.

The tests were carried out at a 66 MW_{th} BFB boiler firing wood chips, bark and saw dust as the main fuels and co-firing green forest residue, chipboard and peat. Peat firing tests were carried out to find out the effect of silicate minerals in ash as well as the effect of sulfur on deposition.

Introduction

Fluidized bed combustion offers the possibility to burn low-grade fuels effectively. Low-grade fuels include wood, wood wastes and agrobiomaterials, as well as demolition wood and recycled fuels. The multifuel combustion is advisable in industry, where the waste fractions having recoverable energy content would otherwise be dumped.

Foster Wheeler Corporation has delivered more than one hundred BFB and CFB boilers for multifuel combustion. Table 1 presents a few examples of boilers firing biofuels and recycled fuels.

Table 1. Commercial plants co-combusting biofuels and recycled fuels.

Plant	Type of boiler	Max. heat capacity MW _{th}	Fuels
• Söderenergi, Sweden	BFB	120	50 % demolition wood 50 % forest residue
• Forssa Energia , Finland	BFB	66	wood, saw dust, forest residue (< 30 % REF) ^{a)}
• IVO, Kauttua, Finland	CFB; hot cyclone	65	20% PDF 20-25 % RDF (1993-1994)
• Enso, Varkaus, Finland	CFB; hot cyclone	150	20 % plastic reject in bark
• Kainuun Voima, Kajaani, Finland	CFB; hot cyclone	240	1% PDF in bark and peat
• Lomma Energi, Sweden	CFB, compact	16	100 % REF
• Hornitex, Germany	CFB, compact /INTREX™	86	50% demolition wood 50 % production reject
• Robbins, USA	CFB	2x74	100 % RDF
• LLV, Lahti, Finland	CFB gasifier connected to coal fired PC boiler	40-70	Recycled fuels, bark, wood wastes

REF Recycled Fuel, PDF Packaging Derived Fuel, RDF Refuse Derived Fuel

^{a)} REF combustion planned

The composition and characteristics of ash vary according to the type of the fuel combusted. In biofuel combustion, the combustion process has occasionally suffered from bed material sintering and fouling of superheaters and other cooling surfaces. The fouling tendency of the fuels is not dependent on their ash content only. The ash content in biofuels is frequently low, but the composition of the fuel may be unfavorable. These characteristics have restricted the wide utilization of the novel biofuels and waste fractions suitable for energy recovery. It is therefore of great importance to study the behavior of biofuels during continuous energy generation to increase detailed understanding of the different ash related problems which the combustion of biofuels may suffer from. In this paper such an approach is presented. The mechanism of fouling was studied in a commercial 66 MW_{th} BFB boiler using sophisticated ash and deposit sampling equipment and novel fuel characterization methods.

The 66 MW_{th} BFB boiler plant generates 19.7 kg/s 61 bar, 510°C steam for power generation and district heat. The main fuel was wood in which other biofuel fractions were mixed, i.e. green forest

residue, chipboard reject and chipboard reject with peat. A temperature-controlled deposit probe was used to simulate the superheater tube and to collect samples of the deposits. Short probe tests were used for studying the deposit at the initial state of formation.

The fly ash amount and particle size distribution was studied during the different test periods. In-situ fly ash samples were collected from the flue gas using an 11 stage low-pressure impactor coupled with a pre-cut cyclone. The fuels used in the tests were analyzed using the novel chemical fractionation analysis technique /Benson 1995, Baxter 1994/.

Test arrangements

The testing facility is a 66 MW_{th} BFB boiler (Figure 1) that uses a mixture of bark, saw dust and wood chips as the main fuel. The fuels are loaded with a front loader on four horizontally moving bottom silos. Each of the four silos are independently feeding the fuel on the drag chain conveyer transporting the fuel blend into two daily silos. From the daily silos the fuel mixture is fed through two feeding points into the furnace of the BFB boiler.

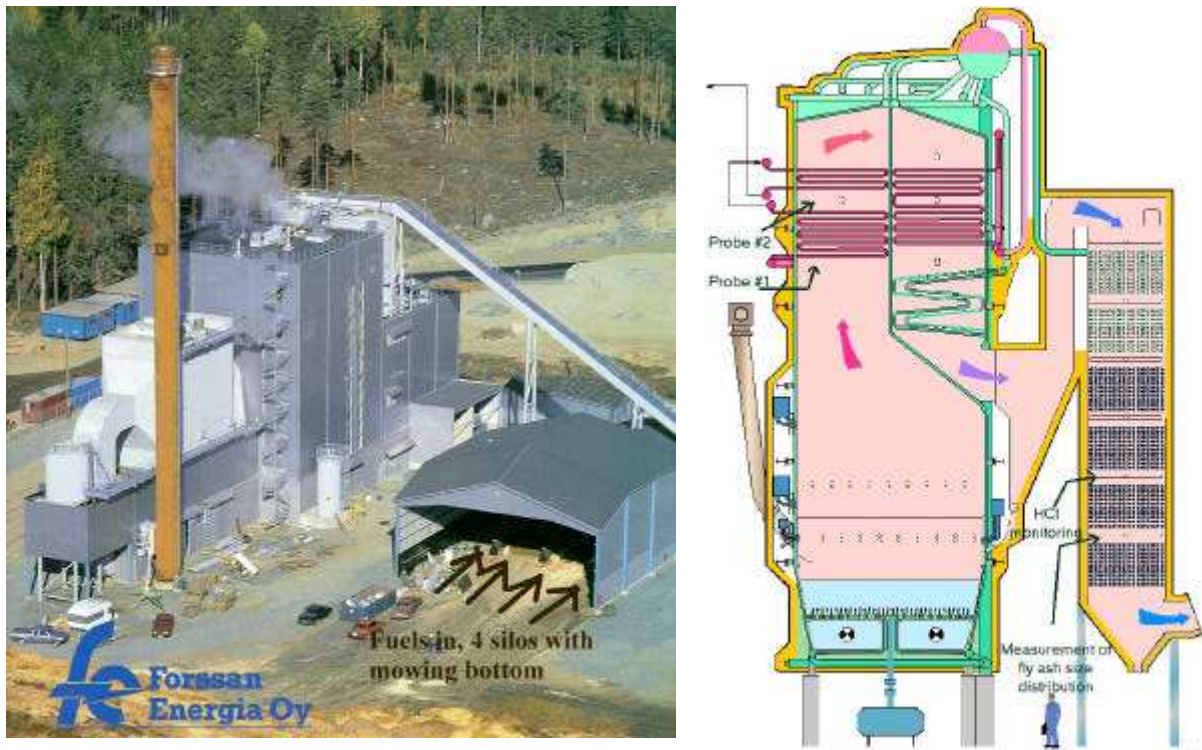


Figure 1. Lay-out of the 66 MW_{th} CFB boiler plant at Forssa /Energia 1996/.

The main fuel mixture was used as the reference fuel. The main fuel was fed via one of the four silos with horizontally moving bottom. The co-combusted fuels were fed via the other silos. The feed rates of the fuels were adjusted by controlling the speed of the bottom conveyers of the feeding silos.

The combustion tests were carried out as one-day combustion tests. Three fuel grades were tested: forest residues (25 %), chipboard waste (24 %) and chipboard with peat (20 % / 28 %). The thermal energy contribution of the co-combusted fuels are presented in parentheses. The load of the boiler

was between 55 % and 67 % of the maximum capacity. The combustion conditions were approximately the following: bed temperature 820°C, flue gas recirculation 8 % , O₂ 4 %, SO₂ <5 ppm and HCl 4-20 ppm.

Fouling of superheaters was tested by using temperature-controlled probes simulating superheater at two temperatures of flue gas, i.e. 850°-900°C and 750°-800°C. The test specimens were made of same material and were the same size as the actual superheater. The deposit formation was monitored by weight increment, and visual examination of the deposits.

Characteristics of fuels

Fuel sampling was carried out at the discharge of each daily silo. The mineral matter of fuels was analyzed from fuel samples ashed at the temperature of 575°C by using XRF. The analysis results of fuels are presented in Table 2.

The green forest residue was harvested during the winter. Due to snow adhering to the fuel its moisture content was high (65%). The chipboard used was reject material from furniture manufacturing and included lined chipboard. Therefore, the titanium content in the two fuel mixtures of chipboard reject was higher than generally in wood fuels. All fuels combusted in the tests could be ranked as low-alkaline and low-chlorine biofuels. Concentrations of Ca and K in the ashes of the fuel mixtures were highest. The total concentrations of Ca and K increased with co-fuels but the fraction of K soluble in ammonium acetate was lowest in the fuel mixture with peat. Besides, the ash of peat had higher Si, Al and Fe contents than that of the wood fuels.

The contents of S, Cl and volatile alkali metals in the fuels are presented in Table 3 as molar ratios. The volatile Na and K were determined on the basis of elements soluble in ammonium acetate. The molar ratio of (K+Na)/(2S+Cl) compares alkali metals with the theoretical amount of alkalis needed to form X₂SO₄ and XCl (X= K or Na) from all sulfur and chlorine. The fuel mixture used in the reference test and in the chipboard test contained an excess of alkalis compared with sulfur and chlorine.

The chipboard in fuel increased the chlorine content so that the molar ratio with sulfur was close to one. During the firing of chipboard/peat mixture, the molar ratio of alkalis compared with that of sulfur and chlorine decreased to such an extent that part of the chlorine escaped the boiler as gaseous HCl. This is well in accordance with earlier experiences /Skrifvars 1997b/. In peat combustion, the ash-forming elements was changed.

Table 2 Analyses of fuel mixtures. Fuel mixtures were ashed at 575°C for semiquantitative XRF analysis.

			Reference test	Forest residue 25 %	Chipboard 24 %	Chipboard 24 % peat 28 %
moisture	as received	%	47.6	51.7	52.3	-
ash	in d.s.	%	1.3	1.5	1.8	4.5
volatiles	in d.s.	%	80.9	78.4	76.8	74.8
C	in d.s.	%	50.9	51.5	50.4	48.9
H	in d.s.	%	6.1	5.9	5.7	4.7
N	in d.s.	%	0.14	0.24	0.54	0.8
S	in d.s.	%	0.03	0.05	0.03	0.06
LHV	in d.s.	MJ/kg	19.06	19.26	19.23	19.03
Na, soluble ^{a)}	In d.s.	µg/g	260	220	340	330
K, soluble ^{a)}	In d.s.	µg/g	800	920	900	640
Na, total	in d.s.	µg/g	360	330	460	470
K, total	in d.s.	µg/g	910	1040	1010	950
Cl total	in d.s.	µg/g	170	170	290	690
Elements in dry fuel by XRF						
Na	in d.s	µg/g	300	260	390	820
K	in d.s	µg/g	780	920	880	1500
Ca	in d.s	µg/g	4100	4700	5000	6700
Mg	in d.s	µg/g	400	480	520	1200
Al	in d.s	µg/g	110	190	350	2900
Fe	in d.s	µg/g	120	250	380	2300
Si	in d.s	µg/g	240	430	730	7100
P	in d.s	µg/g	170	230	250	500
Ti	in d.s	µg/g	28	43	660	920
S	in d.s	µg/g	170	460	200	640
Cl	in d.s	µg/g	(18)	(50)	(160)	(360)
Mn	in d.s	µg/g	220	210	240	240
Zn	in d.s	µg/g	46	60	64	90

a) soluble in ammonium acetate

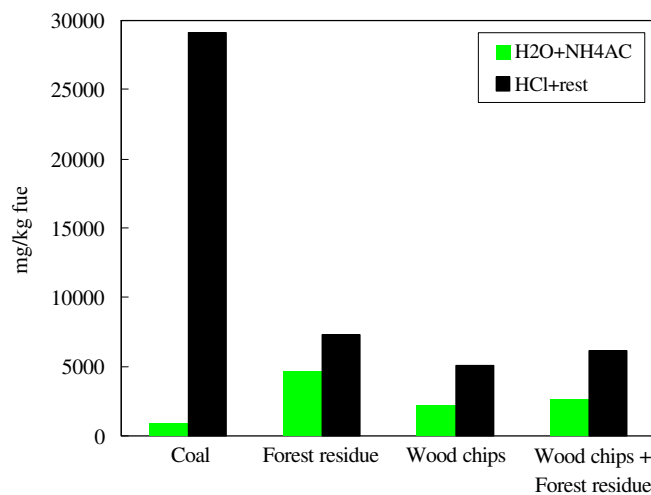
Table 3. The molar ratios of S, Cl and volatile alkalis in fuel mixtures and HCl concentration before the ESP.

	(K+Na)/(2S+Cl) mol/mol	S/Cl mol/mol	Na/K mol/mol	HCl ppm (wet)
Reference fuel mixture 100 %	1.4	2.0	0.5	4
Forest residue 25 %	0.9	3.2	0.4	7
Chipboard 24 %	1.4	1.1	0.6	5
Chipboard 24 % + peat 28 %	0.5	1.0	0.9	20
Peat 100 %	0.1	4.7	0.7	-
Chipboard 100%	0.6	0.2	1.4	-

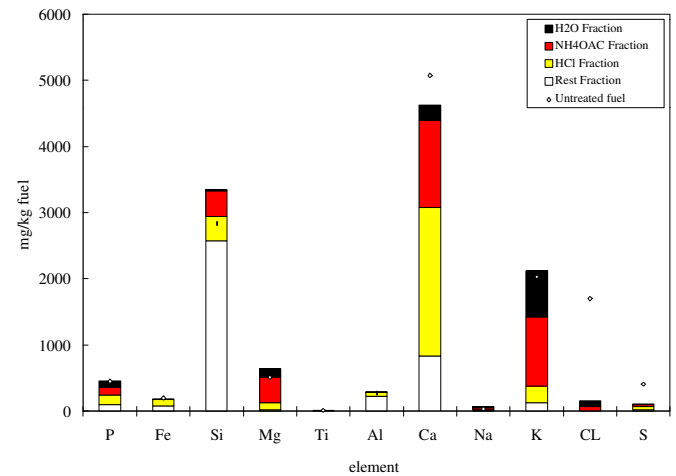
Chemical fractionation analysis

The fuel mixtures were analyzed by a traditional method (Table 3) and by a chemical fractionation method which could provide more information of fuel characteristics relating to ash behavior in fluidized bed combustion. The chemical fractionation analysis was carried out by Åbo Akademi University. In the chemical fractionation, the ash samples were successively leached in water, ammonium acetate and hydrochloric acid. The object of this analysis was to predict the ash behavior in combustion. The prediction method is described elsewhere /Skrifvars 1998/.

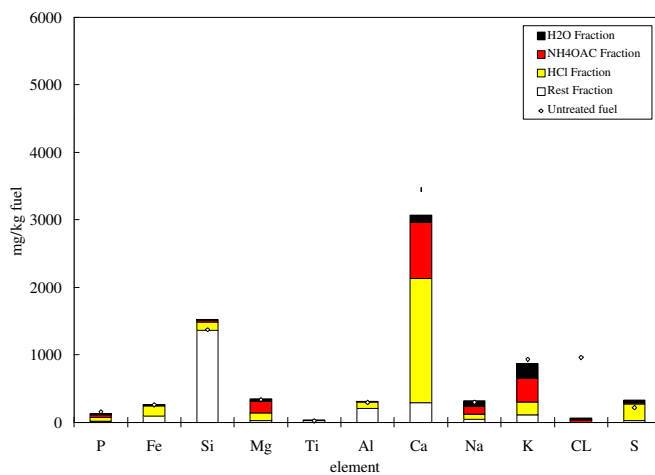
More than half of all ash-forming elements were found in the fraction soluble in HCl acid and in the insoluble rest (Figure 2a). The results of the chemical fractionation analysis of forest residue, reference wood fuel and the mixture of those two fuels are presented in Figure 2 b-d.



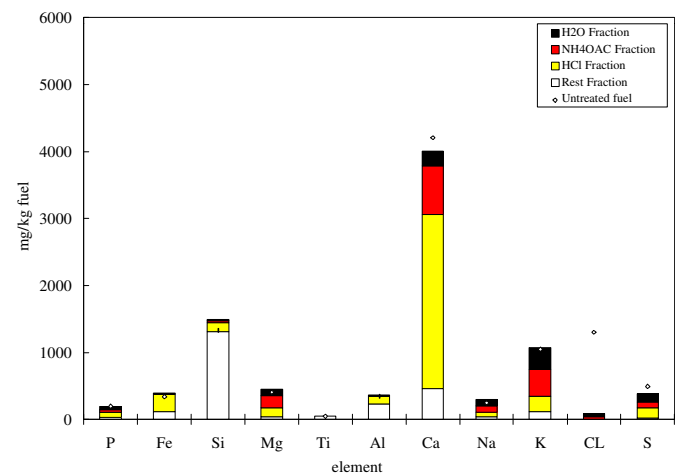
a) chemical fractionation to two phases



b) chemical fractionation of forest residue



c) chemical fractionation of reference wood fuel



d) chemical fractionation of forest residue and wood

Figure 2. Results of chemical fractionation analysis /Skrifvars. 1998/.

The main component in the ashes was Ca. Most of the Ca was analyzed in the fractions of the HCl soluble and the insoluble rest, i.e. mostly in the non-volatile phase during combustion. The other

major component in the ashes was Si, which has a small fraction soluble in the ammonium acetate in forest residue. Potassium, chlorine and sulfur were all mostly in the form of simple salts soluble in water and organically bound compounds soluble in the ammonium acetate. They should form sub-micron particles during combustion, and they are also main components in the volatilized phase /Baxter 1994/.

In general, the results of the chemical fractionation analysis deviates from the XRF results. The biggest variations are found in the concentrations of Cl and Si. Moreover, when fuel is ashed at the laboratory before the XRF analyses, a great deal of Cl is volatilized.

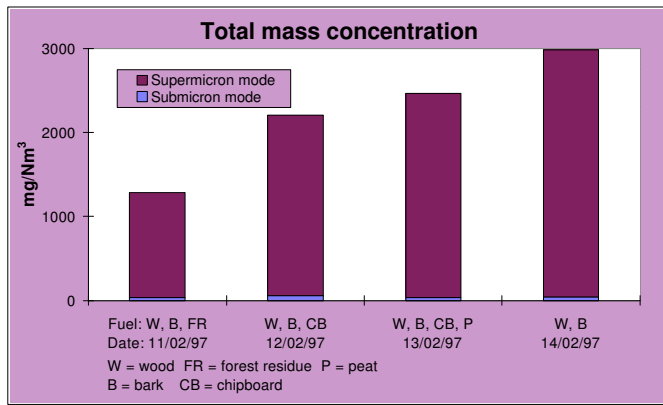
The chemical fractionation analysis was used to predict the amount of minerals forming the so called “fines” which could accelerate deposit formation. The fines particles were assumed to forming elements found in the easily extractable fractions, i.e. in the solutions of water and ammonium acetate at the chemical fractionation analysis. The prediction is presented elsewhere /Skrifvars 1998/.

Measurement of fly ash particle size

The mass-size distributions of the fly ash particles were determined from the samples taken between the last two air pre-heater tube bundles in the convection part of the boiler. The sampling was carried out by an 11-stage, multijet compressible flow Berner-type low-pressure impactor (BLPI) connected to an 8 µm cut-size pre-cyclone. The BLPI method developed at VTT has been described in more detail elsewhere /Kauppinen 1992/. The number-size distributions of particles were measured on-line by an electrical low-pressure impactor (ELPI) at the same location. The elemental mass-size distributions for water soluble ions, i.e. Cl⁻, Na⁺, K⁺, and SO₄²⁻, were derived from the BLPI samples by ion chromatography (IC) and for other elements by inductively coupled plasma-mass spectrometry (ICP-MS).

The total ash concentration in the flue gases for each fuel mixture is presented in Figure 3. The highest total fly ash concentrations were in the combustion of the reference fuel mixture, although the ash content of the fuel mixture was the lowest. The test conditions during the reference test were unstable due to changes of the heat load in the district heating network. The concentration of submicron particles was highest (58 mg/Nm³) in the combustion of the chipboard fuel mixture. Most of the material of sub-micron particles was in a water soluble form.

The highest concentration of Cl in submicron particles was observed during the chipboard combustion. The highest concentrations of Na, K and S in sub-micron particles were found in the reference test. When the concentrations of sulfates were highest in the reference test, the contents of chlorides were highest in the co-combustion of chipboard. Calcium species were not observed in the fine particles, confirming that Ca did not volatilize during combustion /Valmari et.al. 1999 a,b/. The mass and composition distribution of fly ash were similar to those found in other recent studies /Valmari et.al. 1999 b/.



a) total ash concentrations for each fuel mixture

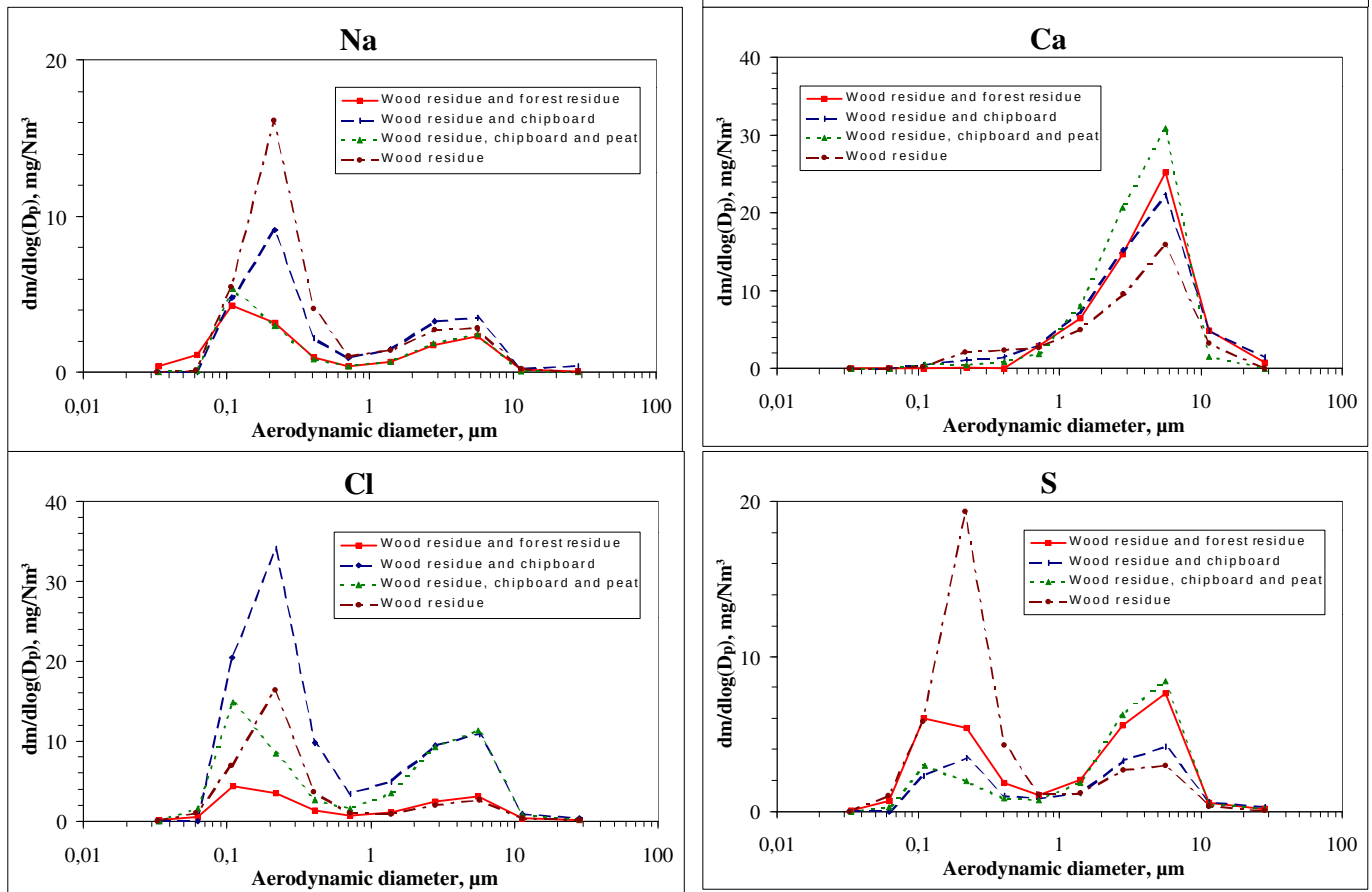


Figure 3. Fly ash mass concentrations and elemental size distributions of K, Na, Ca, Cl and S for particles below 8 μm in the convection of BFB boiler. Elemental size distributions were for water soluble ions. Wood residue was the main fuel and as the reference fuel /Kurkela 1998/.

Fouling of superheaters, probe tests

The composition of ash layers on the deposition probes were different depending on the fuel mixtures. When peat was included in the fuel mixture, the ash layer was looser than the layer in the tests with only wood fuel mixtures. The coarse fly ash particles (>10 μm) were found on the windward side of the probe. They must have been transported on the surface by the inertia

impaction. Loose deposit was collected on the leeward side of the probe by turbulence. Thermophoresis may have promoted the depositing of sub-micron particles on the surface. Heterogeneous gas-solid chemical reactions also affect deposit formation.

The fouling rate on the probes simulating the superheaters is presented in Figure 4. Chipboard co-combustion increased fouling on the surface of the probe. Somewhat surprisingly, forest residue co-combustion gave the lowest fouling rate. This is against the earlier practical experiences from fluidized bed boilers cofiring forest residues. Although the peat co-combustion increased the total amount of ash, the fouling rate decreased. The share of peat decreased the share of wood fuels decreasing the amount of volatilized K. The concentrations of Fe, Al and Si in ash were also changed.

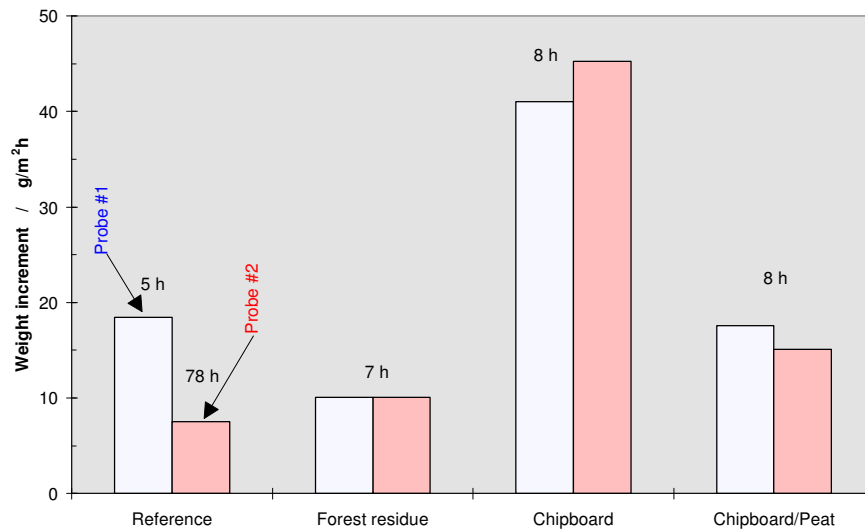


Figure 4. The weight increment of test coupons in deposit probes. The flue gas temperatures at the probes were approximately 850°C (probe #1) and 750°C (probe #2).

The long-term test (78 h) showed that the alkali chloride layer up to 200 µm in thickness had formed. This observation was surprising, because the Cl concentration was low in the fuels. Examples of the deposits on the test probes are presented in Figure 5.

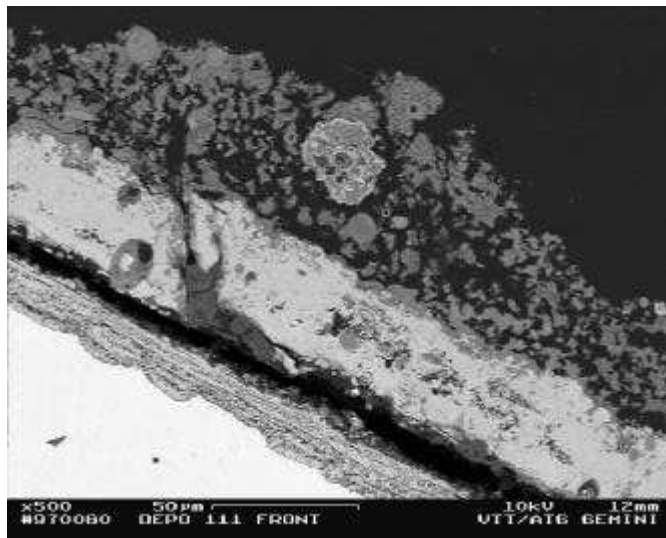
The fly ash analyses are presented in Table 4. The elemental analysis of the fly ash samples was carried out by using an XRF method. The fly ash samples were taken in the discharges of two fields in the electrostatic precipitator (ESP). The coarse particles containing a lot of bed sand were separated in the first field of the ESP (Field I). The fine particles which could be expected to originate from the fuel ash were separated in the second field of the ESP (Field II). The mean size of the particles and the composition were different in these two fly ash fractions. The concentrations of Cl and S were higher in the fly ash from Field II.

Table 4. Fly ash analyses. ESP had two blocks Field I and Field II. Terms of d_{50} , d_{10} , and d_{90} presents the particle size of which 50%, 10% and 90% of particles are smaller

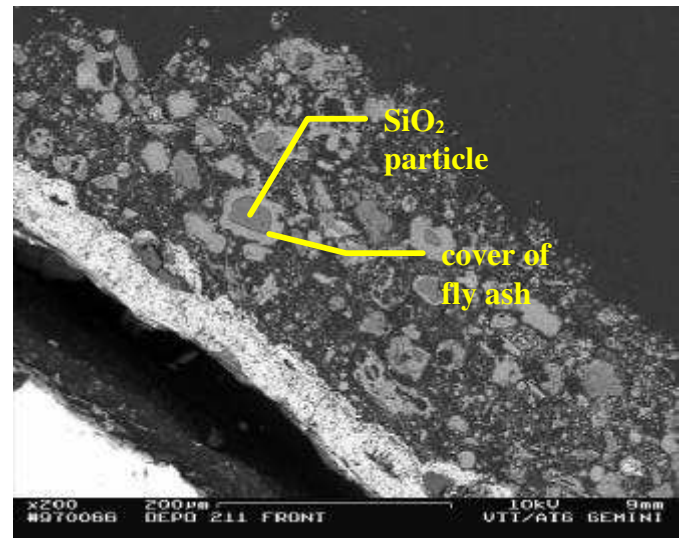
		Reference test		Forest residues 25 %		Chipboard 24 %		Chipboard + peat 20 %/28 %	
		Field I	Field II	Field I	Field II	Field I	Field II	Field I	Field II
S	%	-	-	0.2	3.9	0.8	2.17	0.7	1.6
Cl-	%	0.42	3.5	0.07	2.2	1.5	4.0	1.0	3.2
C	%	4.8	9.4	3.4	5.8	7.3	9.6	3.6	9.4
C-carbonate	%	0.3	1.8	0.11	0.7	1.0	1.6	0.4	1.0
pH		-	-	12	10.3	10.5	-	11.1	-
d_{50}	μm	-	-	360	8.2	240	11	160	24
d_{10}	μm	-	-	140	1.5	4.5	1.6	6.1	2.3
d_{90}	μm	-	-	610	260	550	260	470	220
XRF Analyses									
Na	%	1.6	2.0	1.5	1.4	1.6	1.9	1.7	1.7
K	%	4.0	5.9	3.6	4.4	4.2	5.9	3.5	4.1
Ca	%	10.7	24.3	5.2	21.6	16.5	23.8	13.1	17.4
Si	%	24.2	5.4	30.6	7.5	16.6	5.6	18.3	9.5
S	%	1.4	3.3	0.5	4.4	1.6	2.6	1.6	2.3
Cl	%	1.3	4.8	0.2	2.7	2.8	5.3	2.2	4.6
Al	%	3.8	2.4	5.0	4.1	3.4	2.9	5.5	4.6
P	%	0.8	1.4	0.3	1.6	1.1	1.4	0.9	1.2
Mg	%	1.4	2.5	0.9	2.8	1.9	2.6	2.0	2.3
Fe	%	2.1	3.0	2.3	5.3	2.8	3.7	4.3	5.0
Cr	%	0.03	0.0	0.04	0.07	0.04	0.0	0.03	0.04
Zn	%	0.1	0.3	0.1	0.6	0.2	0.4	0.2	0.3
Ti	%	1.1	2.9	0.4	0.9	2.1	2.6	2.4	3.3
Pb	%	0.02	0.1	0.01	0.13	0.04	0.1	0.04	0.08
V	%	0.01	0.0	0.01	0.03	0.02	0.0	0.02	0.03
Mn	%	0.4	1.3	0.2	1.1	0.8	1.3	0.5	0.8
Ba	%	0.13	0.25	0.1	0.24	0.16	0.22	0.2	0.19

Table 5. Semiquantitative analysis of fuel ash. Fuel mixtures were ashed in laboratory at 575°C.

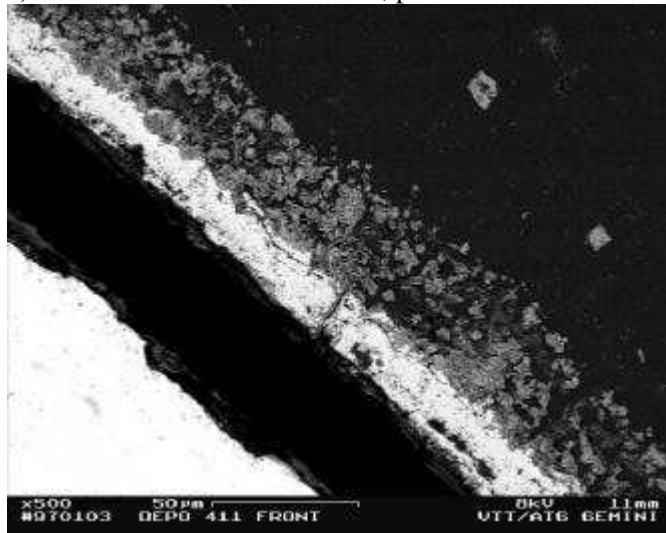
			Reference test	Forest residue 25 %	Chipboard 24 %	Chipboard 24 % peat 28 %
Na	in ash	%	2.4	1.7	2.2	1.8
K	in ash	%	6.1	6.1	4.8	3.3
Ca	in ash	%	31.8	31.6	27.4	14.9
Mg	in ash	%	3.1	3.2	2.9	2.7
Al	in ash	%	0.85	1.2	1.9	6.5
Fe	in ash	%	0.94	1.7	2.1	5.2
Si	in ash	%	1.9	2.9	4.1	15.8
P	in ash	%	1.4	1.5	1.4	1.1
Ti	in ash	%	0.22	0.29	3.7	2.0
S	in ash	%	1.3	3.1	1.1	1.4
Cl	in ash	%	(0.14)	(0.33)	(0.90)	(0.80)
Mn	in ash	%	1.7	1.4	1.3	0.5
C	in ash	%	6.2	4.1	4.6	0.74
Zn	in ash	%	0.36	0.40	0.35	0.20



a) forest residue co-combustion, probe #1



b) chipboard co-combustion, probe #1



c) 78 hours' test probe #1



d) detail of deposit on probe #2 in reference test

Figure 5. Back-scattered electron images of cross-sections of the deposits on the metal test specimens. Metal is shown lightest at the left lower edge and epoxy resin as darkest area /Kurkela VTT/.

Discussion

The combustion of biomass is a process where the composition of fuel and bed material are the primary sources of deposition and agglomeration in a fluidized bed. The process involves release of mineral matter from fuel, formation of ash components, transportation and adherence of the components on the surfaces.

Normally, biofuel contains little ash-forming material, whereas the contents of volatile material are high. The moisture evaporation, volatilization and pyrolysis are all chains of reactions of a biofuel

particle that occur when biofuel is fed into the hot fluidized bed. The simple salts and organically associated metals are prone to be vaporized during combustion and may undergo several transformations, gas-phase chemical reactions and condensation. During char burnout, the mineral grains undergo phase transformations. The minerals in the char matrix may coalesce into one ash particle including separate minerals, they may stay as separate minerals or form agglomerated ash particles. Due to the high share of simple salts and organically bound elements in the fuel, the combustion of wood fuels leads to much smaller sizes of ash particles than the combustion of coal. The main proportion of ash is too fine to stay in the bed.

In a fluidized bed, the fuel is mixed with the bed material. The ash-forming elements of biofuel partly deposit on the surfaces of bed particles, partly form sub-micron fly ash particles remaining in the flue gas and hence being transported with the flue gas into the flue gas channel.

The “coating” behavior of bed particles is regularly detected when firing biomass in a fluidized bed, especially when quartz sand is used as bed material /*Latva-Somppi 1998*/. The ash layer covering the bed particle includes mainly non-volatile ash elements. The quartz core below the “ash coating” reacts with alkalis (K and Na) released during combustion. Usually, calcium and in some cases magnesium and aluminum have also reacted with silicates. Silicates are formed on the free silica (SiO_2) particles of sand. Ca, Mg and Al originate from the surrounding solid ash layer, but the way of transportation of alkali metals is uncertain /*Latva-Somppi 1998*/.

The biofuel ash builds “new” bed material by depositing on the bed particles. Inorganic mixtures formed in bed do not melt at a certain temperature but have a wide temperature range where both the solid phase and the liquid phase are present. Alkali silicates could have a low melting point and may cause sintering of bed. A pure potassium silicate has the first melting temperature at 742°C within the range of K/Si ratio between 0.25 and 0.5 /*Skrifvars 1997*/. Therefore, bed material change must be increased when firing high alkaline biofuels.

Vapors and fly ash particles escaping the furnace may be deposited on the heat transfer surfaces in the boiler through a number of mechanisms: inertial impaction, thermophoresis, condensation and chemisorption. The larger fly ash particles ($<100\mu\text{m}$) deposit via inertial impaction method. Thermophoresis is more important for the particles smaller than a few micrometers. Other possibilities are that gas-phase compounds, e.g. KCl and K_2SO_4 , condense directly on the surfaces, or that gas-phase SO_2 and HCl react with the deposit /*Jokiniemi et.al. 1994*/.

The microstructure of the deposit collected on the probes was analyzed at VTT by using the SEM/EDS method. The deposit closest to the tube surface contained K, S and Ca in concentrations that were higher than in average in fly ash. Also chlorine was concentrated in the deposits close to the metal surface, but not so evenly as K and S. Even pure KCl crystals were observed, as shown in Figure 5d. Heterogeneous gas-solid reactions definitely have a role in deposit formation. The concentration of Ca in the fuel ash is high, but in sub-micron particles it is low, as shown in Figure 3. Hence, Ca deposits mainly as fly ash particles through inertial impaction. Larger fly ash particles appeared separate in the deposits, obviously through inertial impaction after the initial phase of the deposit formation. The condensed matter rich in K, S, O (most likely K_2SO_4) appeared to surround many of the fly ash particles (Figure 5b). SiO_2 particles, surrounded by fly ash, were also deposited through inertial impaction. All the deposits formed in fluidized bed combustion seem to be porous, and apparent slagging was not observed.

However, wood fuel fractions have comparatively low alkali contents and low levels of chlorine. Consequently, combustion of wood fuels leads to a fairly low release of alkali vapors and HCl. The boiler fouling rate should be moderate.

Agrobiofuels, grasses and straws may contain very high levels of alkalis and chlorine compared to wood. Large amounts of alkalis released into the gas phase during combustion result in a high fouling and slagging potential. The deposit formation may cause operational problems with high-alkaline fuels, and also corrosion if the chlorine content in fuel is high.

The chemical fractionation analyses, connected to the thermodynamic calculation of multicomponent multiphase equilibrium, should be a valuable tool to predict fouling. This is needed especially, in the cases of new fuel mixtures when the fouling behavior must be anticipated in terms of fuel properties alone. The method still needs more comprehensive knowledge of the interactions of the ash components converted during the combustion in the CFB boilers /Skrifvars 1998/. The properties of the deposit are also influenced by the operating conditions and the design of the boiler. Therefore, monitoring of fouling in the commercial boilers is required in order to verify the phenomenon in practice.

Conclusions

Biofuel combustion results in the formation of gaseous alkaline sulfates and chlorides which can condense on the heat exchanger surfaces. The depositing rate is related to the concentrations of alkalis, sulfur and chlorine in biofuel.

The deposit layer was porous. Similar ash particles as in the bed were deposited via impaction mechanism on the test probes. The deposits consisted mainly of solid particles. The finer particles should have deposited by thermophoresis, condensation, and chemical reactions. However, at operating temperature, it could contain a liquid phase as an inhomogeneous mixture with solid particles.

Against expectations based on the experiences from other boilers, the co-combustion of forest residue caused a lower depositing rate on the test probe than in the reference test. The results of the chemical fractionation analyses and the traditional fuel analyses were similar in the fuel mixtures of the reference test and the forest residue test. Nevertheless, the fly ash concentration of the flue gas in the forest residue test was only half of that in the reference test. The concentrations of K, Na and Cl soluble in water were also lowest in sub-micron particles of fly ash in the co-combustion of forest residue.

The highest fouling rate on the probe surface was detected in the co-combustion of chipboard with reference wood fuel. Chipboard was lined, and water-proof treated. The concentration of Na soluble in the ammonium acetate was higher than in the reference fuel and in the mixture of forest residue. Also the Cl concentration was higher. The molar ratio of S/Cl was 1 for chipboard mixture compared with 3.2 for the mixture of forest residue. The sub-micron particles soluble in water have a high

concentration of K, Na and Cl in the co-combustion test of chipboard. It is believed that the increase of Na and Cl concentrations in the fuel was the reason for the higher fouling rate when firing chipboard.

The addition of peat to the fuel mixture of chipboard decreased the depositing rate. The molar ratio $(K+Na)/(2S+Cl)$ of the fuel mixture was decreased to 0.5. The amount of sulfur and chlorine was twice the amount needed to form sulfates and chlorides from all volatile alkalis in the fuel mixture. This was also indicated by an increased HCl concentration.

The fly ash measurements carried out gave the essential information which was used to clarify the morphology and the formation mechanisms of the deposits. On the other hand, the results of the chemical fractionation analyses correlate reasonably with the results of traditional bulk analyses. Further evaluations are needed for the multicomponent, multiphase equilibrium calculations. Thereafter, the prediction method should be the tool required to estimate the fouling characteristics when firing biofuel mixtures with new compositions. It must be remembered that also the operating conditions of the combustion process must be adjusted for biofuel combustion.

The economical operating of boilers requires correct, advance knowledge of biofuels, to optimize the combustion. The chemical and physical characteristics must be interpreted to a suitable form for the boiler operators. The study in this area is emphasized.

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